

The trade-off: LPDDR max capacities are typically 8, 16, or 32GB on a system (though a few high-end models now offer 64GB LPDDR5X, like some HP EliteBooks). DDR5, on the other hand, can go higher (gaming laptops and workstations can have 64GB or more if they have two DIMM slots). If you need huge memory (database work, VMs, heavy scientific computing), look for DDR5 machines with upgradeable slots. For most users, 16GB is the sweet spot. Note that DDR5, while very fast, uses a higher voltage (1.1V vs ~0.5–1.05V for LPDDR5X) and can be less power-efficient – meaning slightly shorter battery life – but the difference is not huge unless you're pushing the RAM constantly.

- **Dual-channel & “lanes”:** The term “lanes” in this context often refers to memory channels or the width of the memory bus. Dual-channel memory (two sticks or a dual 64-bit bus) effectively doubles throughput versus single-channel. This is crucial for integrated GPU performance – an iGPU uses system RAM, so more bandwidth = better graphics. For instance, a test on an AMD Ryzen laptop found that adding a second RAM stick (going from single to dual-channel) boosted graphics performance by an eye-popping 65%! Gamers and creators should ensure dual-channel config (most laptops with soldered LPDDR are inherently multi-channel; the risk is mainly in laptops that ship with one removable stick – avoid a single 8GB stick setup if possible, opt for 2×8GB). With a discrete GPU, single vs dual-channel RAM still affects overall performance, but much less so (CPU tasks see a smaller uplift, and the GPU has its own VRAM). Bottom line: if you buy a laptop with upgradeable DDR5, use two identical sticks for best performance. If you buy an ultrabook with LPDDR, you're already getting a wide memory bus – just choose an adequate amount since you can't upgrade later.
- **Optimal capacities by budget:** 8GB RAM is entry-level and really only advisable for basic use (browsing with few tabs, editing documents). Many budget laptops around ₹40,000–50,000 still come with 8GB; they work for students or light use, but can feel limiting with Windows 11 and multitasking. 16GB RAM is the comfort zone for most people in 2025 – smooth multitasking, enough for heavy web use, some light photo/video editing, and moderate gaming. Laptops in the ₹50k–80k range often have 16GB by default. 32GB RAM is great for power users: running VMs, large Excel datasets, professional content creation, etc. You'll typically find 32GB in higher-end laptops ₹1 lakh and above (or configurable as an upgrade). Only niche users need 64GB+ (3D animation professionals, large scientific simulations, etc., or those who want to seriously future-proof). For storage, SSD is a must – thankfully, almost all new laptops use NVMe SSDs now. 256GB SSD is bare minimum (found in some entry models – you might want to immediately expand or use cloud storage). 512GB is a comfortable minimum for most, allowing room

RAM & STORAGE TIPS FOR LAPTOPS

When buying, check if the RAM is soldered. If yes (common with LPDDR5X ultrabooks), choose a configuration with enough RAM for your future needs, since you can't add more. If the RAM is socketed (common in gaming and business laptops), you can opt for 8GB or 16GB now and plan to upgrade later (just ensure there's an extra slot or the manual says “upgradeable to X GB”). For storage, it's often easy to upgrade SSDs later, but opening the laptop might void warranty in some cases – if you're not comfortable doing that, get the larger SSD up front.

As a guideline by budget in India: under ₹50k, expect 8GB/256GB; ₹50–70k range, aim for 16GB/512GB; ₹70k–1L, 16GB/1TB or 32GB/1TB on premium models. High-end laptops (>₹1L) often come with 32GB/1TB standard. Always prefer the faster memory type if given an option – e.g., DDR5 over DDR4, or LPDDR5X over LPDDR4x – for better efficiency and longevity (and because older types indicate an outdated model). And remember, more RAM and SSD can often improve a laptop's usability more than a slightly faster CPU, especially if you multitask or store lots of data.

for OS, applications, and plenty of files. 1TB is ideal if you work with large media files or games (each modern game can be 50–100GB). If your budget allows, go 1TB, or ensure the laptop has an extra SSD slot for future expansion.

- **Storage specs (PCIe Gen3 vs 4 vs 5):** Newer SSDs advertise crazy speeds (PCIe Gen4 x4 drives can reach ~7 GB/s, Gen5 even higher). In practice, for everyday use (booting Windows, launching apps, loading game levels) you won't notice a big jump beyond SATA SSD speeds except in specific scenarios. The main thing: get an NVMe SSD, any modern one will be fine. Do note that some thin-and-light laptops, especially cheaper ones, might use a narrower interface (like PCIe x2) to save power/cost – this halves the bandwidth (~1.5–2 GB/s instead of ~3–4 GB/s). It's not a deal-breaker unless you routinely transfer huge files. If you do a lot of 4K video editing off the internal drive, favor laptops that explicitly mention PCIe 4.0 x4 SSDs. For most, a Gen3 x4 SSD is already plenty fast. Avoid eMMC storage (found only on very low-end notebooks or tablets, eMMC is much slower flash storage). Also, 2025 laptops rarely have hard drives except some bulky gaming models that might offer a secondary 2.5" bay for cheap mass storage – SSD should be your primary drive for speed.